DEXTROUS

6 DOF Robotic Arm with Locomotion

1) TOY MOTOR SPECS:

These are standard '130 size' DC hobby motors. They come with a wider operating range than most toy motors: from 4.5 to 9V DC instead of 1.5-4.5 V. This range makes them perfect for controlling with an Arduino or ESP 32 where you are more likely to have 5 or 9 V available than a high current 3V setting. They'll fit in most electronics with 130-size motors installed and two breadboard-friendly wires already soldered on for fast prototyping.

Technical Details:
Operating Temperature: -10°C ~ +60°C
Rated Voltage: 6.0VDC
Rated Load: 10 g*cm
No-load Current: 70 mA max
No-load Speed: 9100 ±1800 rpm
Loaded Current: 250 mA max
Loaded Speed: 4500 ±1500 rpm
Starting Torque: 20 g*cm
Starting Voltage: 2.0
Stall Current: 500mA max
Body Size: 27.5mm x 20mm x 15mm
Shaft Size: 8mm x 2mm diameter
Weight: 17.5 grams

Datasheet:
https://www.farnell.com/datasheets/1863913.pdf

2) SERVO MOTORS SPECS:

MG995 High-Speed Servo

- Weight: 55 g
- Dimension: 40.7 x 19.7 x 42.9 mm approx.
- Stall torque: 8.5 kgf·cm (4.8 V), 10 kgf·cm (6 V)
- Operating speed: 0.2 s/60o (4.8 V), 0.16 s/60o (6 V)
- Operating voltage: 4.8 V a 7.2 V
- Dead band width: 5 µs
- Stable and shock proof double ball bearing design
- Temperature range: 0 oC – 55 oC

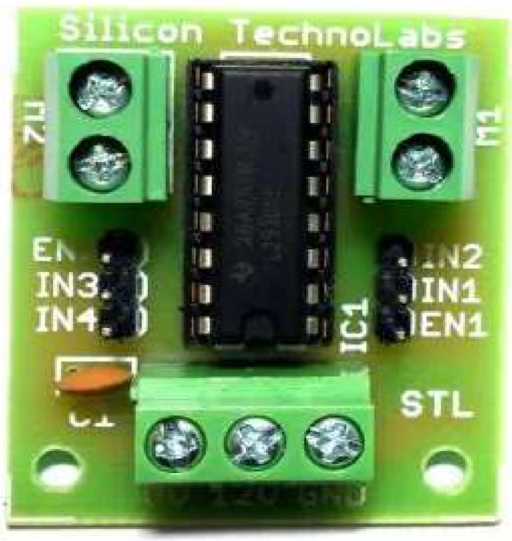
Datasheet:
https://www.electronicoscaldas.com/datasheet/MG995_Tower-Pro.pdf

Calculations:
1 kgf.cm ≈ 13.887 oz-in
So, if each MG995 servo motor provides 8.5 kgf.cm of torque:
8.5 kgf.cm * 13.887 oz-in/kgf.cm ≈ 118.0485 oz-in

1 oz = 28.3 grams,
Hence, 118.04 * 28.3 = 3340.55 grams
Therefore, one servo motor can hold approx 3 kg load.
Total torque = 6 motors * 118 oz-in/motor = 708 oz-in.

Mechanical advantage and efficiency play crucial roles. Let's assume a reasonable mechanical advantage and efficiency, and let's say the robotic arm can lift approximately 5 kg (11 lbs) with this torque.

3) L293D Motor Driver:



L293D Motor Driver Module

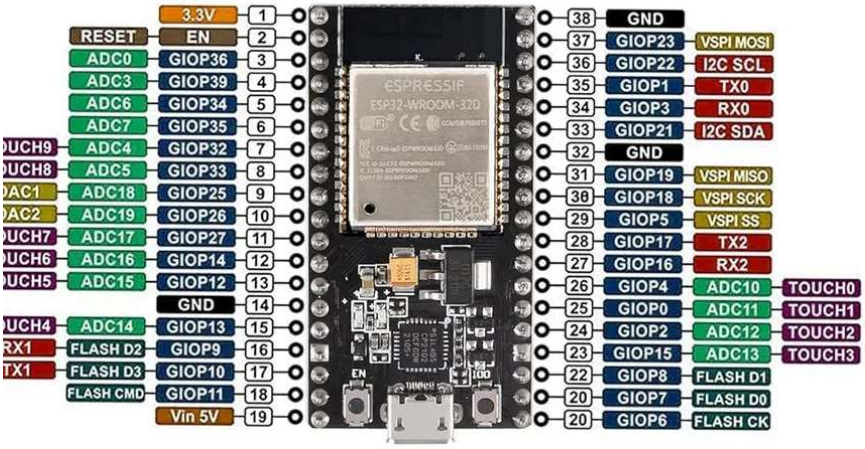
Wide Supply-Voltage Range: 4.5 V to 36 V
Separate Input-Logic Supply
Internal ESD Protection
High-Noise-Immunity Inputs
Output Current 600 mA Per Channel
Peak Output Current 1.2 A Per Channel
Output Clamp Diodes for Inductive Transient Suppression
Operation Temperature 0°C to 70°C.
Automatic thermal shutdown is available

Working of L293D Motor Driver IC:

There are 4 input pins for direction control in L293d. Pin 2,7 (1A and 2A) on the left side and pin 15,10 (3A and 4A) on the right of the IC. The left-side input pins regulate the rotation of the motor connected across the left end and the right-side input pins regulate the motor on the right side. The motors are rotated based on the inputs provided across the input pins as HIGH or LOW signals.
For example, a motor is connected on the left side output pins (pins 3,6). To control this motor, we have to provide an input logic to pin 2,7 (1A,2A).
Pin 2 = HIGH and Pin 7 = LOW | Clockwise Direction
Pin 2 = LOW and Pin 7 = HIGH | Counterclockwise Direction
Pin 2 = LOW and Pin 7 = LOW | Idle (No rotation)
Pin 2 = HIGH and Pin 7 = HIGH | Idle (No rotation)
In a similar manner, we can control the motor on the right side connected to the pin (11,14). We need to provide HIGH and LOW input signals across pins (10,15).
Pin 10 = HIGH and Pin 15 = LOW | Clockwise Direction
Pin 10 = LOW and Pin 15 = HIGH | Counterclockwise Direction
Pin 10 = LOW and Pin 15 = LOW | Idle (No rotation)
Pin 10 = HIGH and Pin 15 = HIGH | Idle (No rotation)

Datasheet:
https://pdf.datasheetcatalog.com/datasheet/stmicroelectronics/1330.pdf

4) ESP-32 Microcontroller:



Main processor: Tensilica Xtensa 32-bit LX6 microprocessor
Cores: 2 or 1 (depending on variation)
All chips in the ESP32 series are dual-core except for ESP32-S0WD, which is single-core.
Clock frequency: up to 240 MHz
Performance: up to 600 DMIPS
Ultra low power co-processor: allows you to do ADC conversions, computation, and level thresholds while in deep sleep.
Wireless connectivity:
Wi-Fi: 802.11 b/g/n/e/i (802.11n @ 2.4 GHz up to 150 Mbit/s)
Bluetooth: v4.2 BR/EDR and Bluetooth Low Energy (BLE)
Internal memory:
ROM: 448 KiB
For booting and core functions.
SRAM: 520 KiB
For data and instruction.
RTC fast SRAM: 8 KiB
For data storage and main CPU during RTC Boot from the deep-sleep mode.
RTC slow SRAM: 8 KiB
For co-processor accessing during deep-sleep mode.
eFuse: 1 Kibit
Of which 256 bits are used for the system (MAC address and chip configuration) and the remaining 768 bits are reserved for customer applications, including Flash-Encryption and Chip-ID.
Embedded flash:

Flash connected internally via IO16, IO17, SD_CMD, SD_CLK, SD_DATA_0 and SD_DATA_1 on ESP32-D2WD and ESP32-PICO-D4.
0 MiB (ESP32-D0WDQ6, ESP32-D0WD, and ESP32-S0WD chips)
2 MiB (ESP32-D2WD chip)
4 MiB (ESP32-PICO-D4 SiP module)
External flash & SRAM: ESP32 supports up to four 16 MiB external QSPI flashes and SRAMs with hardware encryption based on AES to protect developers' programs and data. ESP32 can access the external QSPI flash and SRAM through high-speed caches.
Up to 16 MiB of external flash are memory-mapped onto the CPU code space, supporting 8-bit, 16-bit, and 32-bit access. Code execution is supported.
Up to 8 MiB of external flash/SRAM memory are mapped onto the CPU data space, supporting 8-bit, 16-bit and 32-bit access. Data-read is supported on the flash and SRAM. Data write is supported on the SRAM.
ESP32 chips with embedded flash do not support the address mapping between external flash and peripherals.
Peripheral input/output: Rich peripheral interface with DMA that includes capacitive touch, ADCs (analog-to-digital converter), DACs (digital-to-analog converter), I²C (Inter-Integrated Circuit), UART (universal asynchronous receiver/transmitter), CAN 2.0 (Controller Area Network), SPI (Serial Peripheral Interface), I²S (Integrated Inter-IC Sound), RMII (Reduced Media-Independent Interface), PWM (pulse width modulation), and more.
Security:
IEEE 802.11 standard security features all supported, including WFA, WPA/WPA2 and WAPI
Secure boot
Flash encryption
1024-bit OTP, up to 768-bit for customers
Cryptographic hardware acceleration: AES, SHA-2, RSA, elliptic curve cryptography (ECC), random number generator (RNG)

Datasheet:
https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf

5) Playstation-4:

The originally released 420 GB HDD PS4s had manufacturer serial numbers of the form CUH-10XXA; a minor modification with a different form of Wi-Fi Microstrip antenna was registered in mid 2014 as part numbers CUH-11XXA.
In 2015, the CUH-12 series as variants CUH-1215A and CUH-1215B (with 500GB and 1TB storage respectively) were certified in the US by the FCC. Differences between the CUH-11 and CUH-12 series included a reduction in rated power from 250W to 230W, a reduction in weight from 2.8 to 2.5 kg, and physical buttons. The CUH-12xx series is also referred to as the "C chassis" variant of the PS4.